

JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIOS & IDENTITIES 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Trigonometric Ratio And Identites — Standard Calculations & Problem Archetypes	11 PYQs	Quantitative Analysis
Standard	Trigonometric Ratio And Identites — Core Principles & Governing Laws	1 PYQs	Core Principle
Standard	Trigonometric Ratio And Identites — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Trigonometric Ratio And Identites — Standard Calculations & Problem Archetypes	$\sin(x+\frac{\pi}{6})+\cos(x+\frac{\pi}{6}), \quad \sin\frac{5\pi}{12}\sin\frac{\pi}{12}, \quad \tan1^\circ\tan2^\circ\tan3^\circ\dotso\tan89^\circ$
Trigonometric Ratio And Identites — Core Principles & Governing Laws	$y=\sqrt{2+\sqrt{2+\sqrt{2+2\cos8\theta}}}, \quad \sqrt{2+\sqrt{2+\sqrt{2+2\cos8\theta}}} =$
Trigonometric Ratio And Identites — Applied Cases, Reactions & Traps	$\cos5\pi=\cos(\pi+4\pi)=-1,\cos4\pi=1, \quad \sin2\pi=0,\sin(-2\pi)=-\sin2\pi=0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Ratio And Identities — Standard Calculations & Problem Archetypes (High Yield — 11 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If α and β are acute angles such that $\alpha - \beta$ and $\alpha + \beta$ satisfy the equation $\tan^2 2\theta - 4\tan\theta + 1 = 0$, $t = \dots$ " [Ans: A]
- [2026 Shift] Q#2 (MCQ): "The maximum value of $\sin(x + (\pi/6)) + \cos(x + (\pi/6))$ is attained at $x = \dots$ " [Ans: D]
- [2025 Shift] Q#4 (MCQ): "The minimum value of $1 - \sin x$ is..." [Ans: N/A]

3.2 Trigonometric Ratio And Identities — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Ratio And Identities.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Ratio And Identities PYQs.

Representative Source PYQs:

- [2022 Shift] Q#3 (MCQ): " $\sqrt{(2 + \sqrt{2} + \sqrt{2} + 2\cos 8\theta)} = \dots$ " [Ans: N/A]

3.3 Trigonometric Ratio And Identities — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Which of the following is not correct?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If α and β are acute angles such that $\alpha - \beta$ and $\alpha + \beta$ satisfy the equation $\tan^2\theta - 4\tan\theta + 1 = 0$, then α and β are respectively,

- (A) $45^\circ, 30^\circ$ [Correct]
- (B) $30^\circ, 45^\circ$
- (C) $30^\circ, 60^\circ$
- (D) $60^\circ, 45^\circ$

Correct Answer: A

Solution Analysis: Put $x = \tan\theta$. The given equation is $\tan^2\theta - 4\tan\theta + 1 = 0$ So, $x^2 - 4x + 1 = 0$ Using the quadratic formula, $x = \frac{4 \pm \sqrt{(16-4)}}{2}$ $x = \frac{4 \pm 2\sqrt{3}}{2}$ $x = 2 \pm \sqrt{3}$ Now, $\tan 15^\circ = 2 - \sqrt{3}$ and $\tan 75^\circ = 2 + \sqrt{3}$ So the two angles satisfying the equation are 15° and 75° Here the angles are $\alpha - \beta$ and $\alpha + \beta$. Since $\alpha + \beta$ is greater than $\alpha - \beta$, $\alpha - \beta = 15^\circ$ and $\alpha + \beta = 75^\circ$ Adding both equations, $2\alpha = 90^\circ$ $\alpha = 45^\circ$ Now, $\alpha - \beta = 15^\circ$

Benchmark Problem 2: KCET 2025 (Q#3)

Which of the following is not correct?

- (A) $\cos 5\pi = \cos 4\pi$
- (B) $\sin 2\pi = \sin(-2\pi)$
- (C) $\sin 4\pi = \sin 6\pi$
- (D) $\tan 45^\circ = \tan(-315^\circ)$

Correct Answer: None

Solution Analysis: Let's check each identity: Option A: $\cos 5\pi = \cos(\pi + 4\pi) = -1$, $\cos 4\pi = 1$ These are not equal, so A is false. Option B: $\sin 2\pi = 0$, $\sin(-2\pi) = -\sin 2\pi = 0$ Option C: $\sin 4\pi = 0$, $\sin 6\pi = 0$ Option D: $\tan 45^\circ = 1$, $\tan(-315^\circ) = \tan(45^\circ) = 1$ Only Option A fails, so the incorrect statement is Option A.

Benchmark Problem 3: KCET 2022 (Q#3)

$\sqrt{(2 + \sqrt{2} + \sqrt{2} + 2\cos 8\theta)} =$

- (A) $\sin 2\theta$
- (B) $2\cos\theta$
- (C) $2\sin\theta$
- (D) $2\cos(\theta/2)$

Correct Answer: None

Solution Analysis: Let $y = \sqrt{(2 + \sqrt{2} + \sqrt{2} + 2\cos 8\theta)}$ We know, $1 + \cos 2A = 2\cos^2 A$ Therefore, $y = \sqrt{(2 + \sqrt{2} + \sqrt{2} \cdot (1 + \cos 8\theta))} = \sqrt{(2 + \sqrt{2} + \sqrt{2} \cdot 2\cos^2 4\theta)} = \sqrt{(2 + \sqrt{2} + 2\cos 4\theta)} = \sqrt{(2 + \sqrt{2} \cdot 2\cos^2 2\theta)} = \sqrt{(2 + 2\cos 2\theta)} = \sqrt{(2 \cdot 2\cos^2 \theta)} = 2\cos\theta$